Diary Data Analysis

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Contents

0) Tables of Group Numbers (Important for Sensitivity Analyses!) **2**

1) Impact of Training on Child Rated Antisocial Behaviour at Two Weeks - Sensitivity **4**

Boostrapped Linear Regression model 4

Results 4

Confidence Intervals 4

Bootstrap Diagnostics 6

2) Impact of Training on Teacher Rated Antisocial Behaviour at Two Weeks - Sensitivity Analyses **8**

Bootstrapped Linear Regression 8

Results 9

Confidence Intervals 9

Bootstrap Distribution 10

0) Tables of Group Numbers (Important for Sensitivity Analyses!)

I am running sensitivity analyses as follows:

I will write code to set the conditions for sensitivity analyses, and apply these to the data. I will then run the script with each of these conditions and save each one as a separate PDF.

Printing tables of Ns for each analyses so can confirm analysis is being run no correct data. Note the Ns should be as follows:

- Main analyses
  - Child asb = 82
  - Teach asb = 102
- Sensitivity Analyses 1 & 2 (controlling for training group size and number of sessions completed)
  - These analyses are conducted on the original sample; therefore, the sample sizes (Ns) are the same as in the main analysis.
  - They are included at the end of the main analysis PDF alongside the main analyses.
- Sensitivity 3 (child outlier removed)
  - Child asb = 81
- Sensitivity 4 (only participants with full data)
  - Child asb = 49
  - Teach asb = 61
- Sensitivity 5 (participants with any failed sessions sessions removed)
  - Child asb = 37
  - Teach asb = 49
- Sensitivity 6 (removing any children whose teachers changed between screening and diaries)
  - Child asb = 71
  - Teach asb = 91

Analysis reported here:

```
## [1] "Remove_kids_with_teacher_change"
```

Table 1 : child ASB

Training_Condition	n
CONTROL	38
TRAIN	33

```
## [1] "n participants total = "
```

```
## [1] 71
```

Table 2: teach ASB

Training_Condition	n
CONTROL	46
TRAIN	45

```
## [1] "n participants total = "
```

```
## [1] 91
```

1) Impact of Training on Child Rated Antisocial Behaviour at Two Weeks - Sensitivity

Boostrapped Linear Regression model

This indicates that the initial behaviour score contributes significantly, but the training condition does not. The confidence intervals are wide for the training, and the bias is moderate.

```
# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb <- boot(data = child_asb_d1d2_wide,
                     statistic = boot_coef_c_asb,
                     R = 5000)
```

Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_asb_d1d2_wide, statistic = boot_coef_c_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##      original      bias    std. error
## t1* 4.3483847 -0.27771059    2.673050
## t2* 0.6749262  0.01115234    0.114661
## t3* 0.8622679  0.12082778    3.645092
```

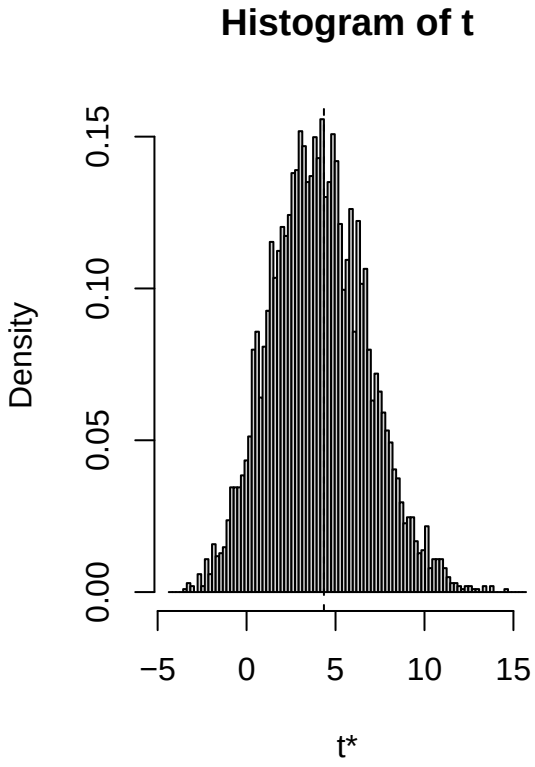
Confidence Intervals

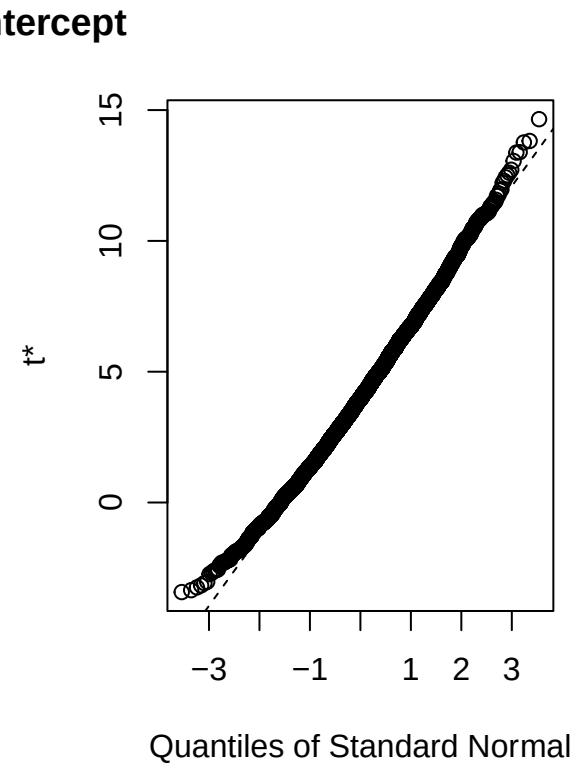
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   ( 0.228, 11.148 )
## Calculations and Intervals on Original Scale
```

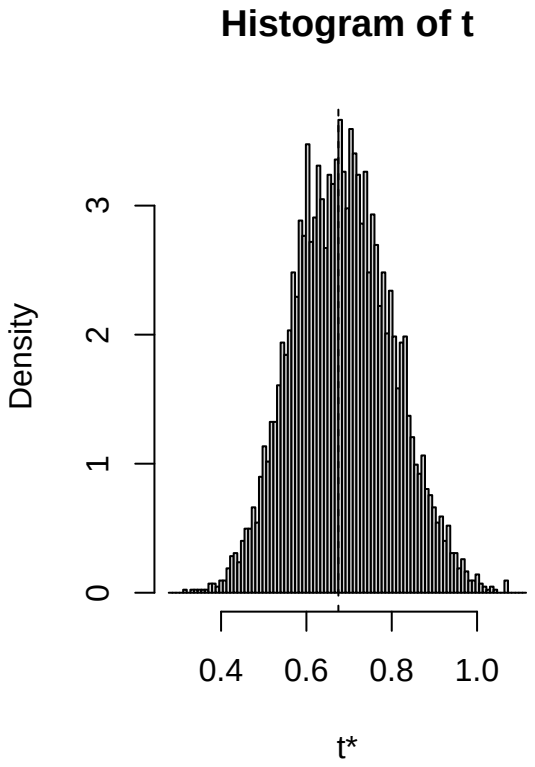
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.4287,  0.8771 )
## Calculations and Intervals on Original Scale

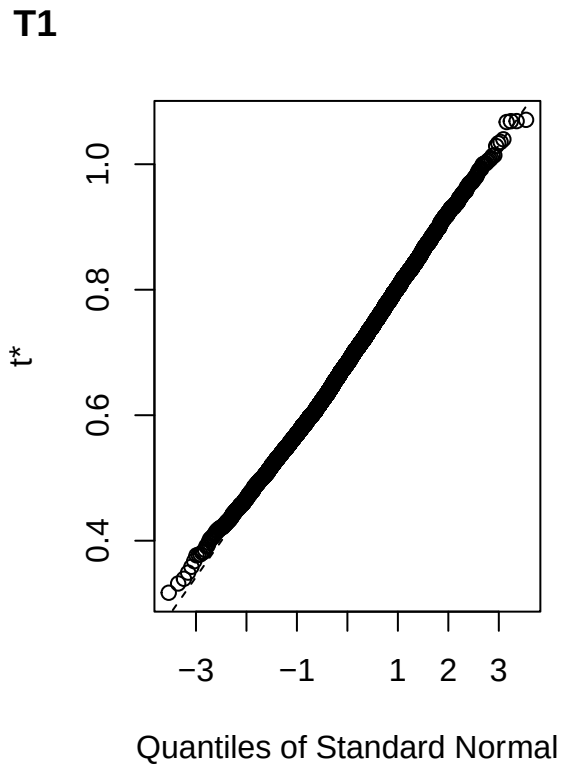
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-6.2471,  7.9545 )
## Calculations and Intervals on Original Scale
```

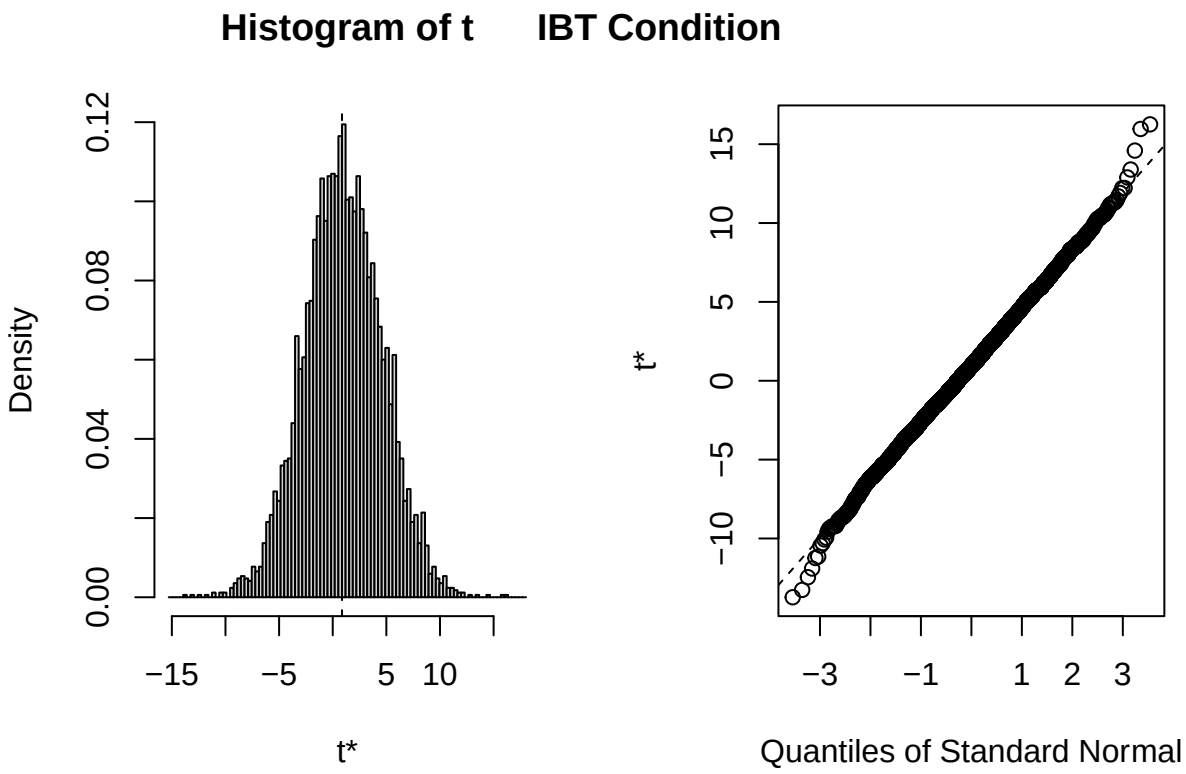
Bootstrap Diagnostics










Histogram of t IBT Condition
Density
0.00
0.04
0.08
0.12
−15
−5
5
10
t*
t*
−10
−5
0
5
10
15
−3
−1
1
2
3
Quantiles of Standard Normal

2) Impact of Training on Teacher Rated Antisocial Behaviour at Two Weeks - Sensitivity Analyses

Bootstrapped Linear Regression

```
# non-parametric case bootstrap: https://journal.r-project.org/articles/RN-2002-017/RN-2002-017.pdf
# argued to be better  for heteroskedastic data: https://library.virginia.edu/data/articles/bootstrappi
boot_coef_t_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
## Boot function: Generates R bootstrap replicates of a statistic applied to data.
set.seed(123)  # For reproducibility
boot_results_t_asb <- boot(data = teach_asb_d1d2_wide,
                     statistic = boot_coef_t_asb,
                     R = 5000)
```

Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_asb_d1d2_wide, statistic = boot_coef_t_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original       bias    std. error
## t1* 11.2906950 -0.008327100  2.25169172
## t2*  0.4649527  0.003608205  0.08855503
## t3* -2.9809817 -0.039350134  2.86208905
```

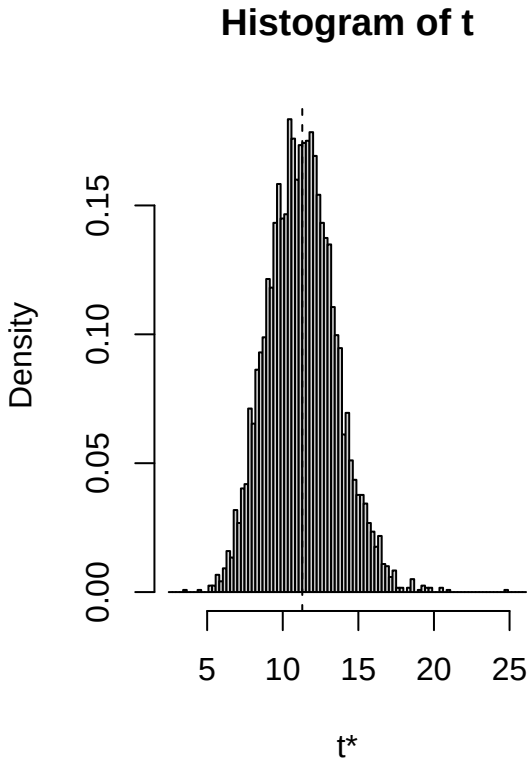
Confidence Intervals

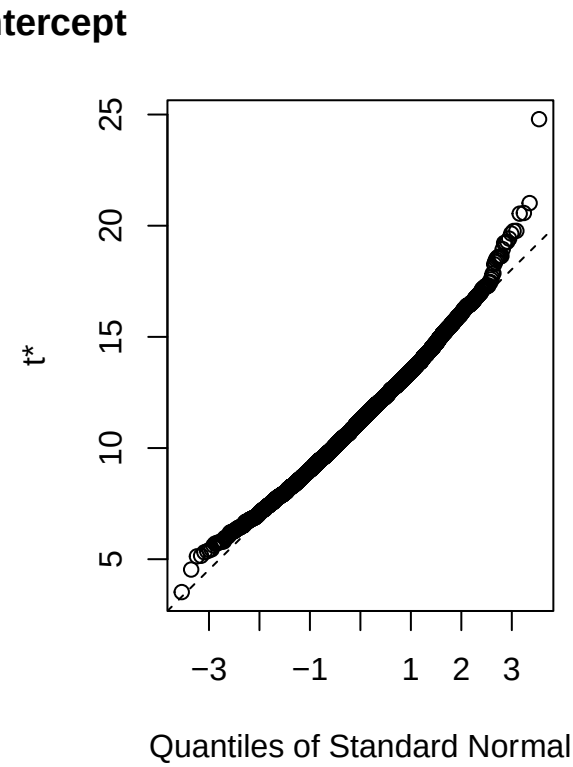
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   ( 7.43, 16.40 )
## Calculations and Intervals on Original Scale

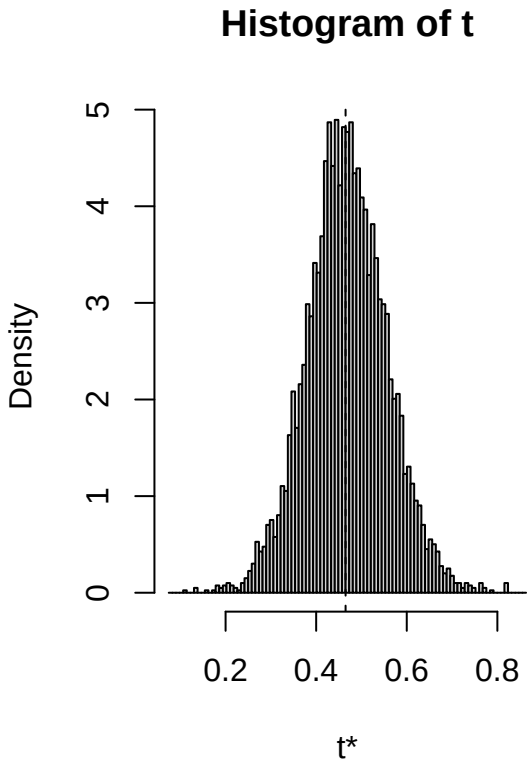
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.2785,  0.6324 )
## Calculations and Intervals on Original Scale

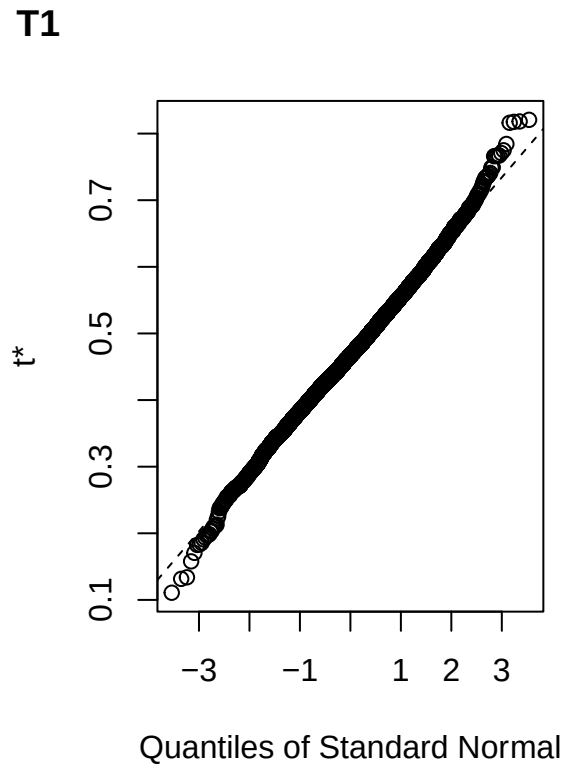
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-8.491,  2.651 )
## Calculations and Intervals on Original Scale
```

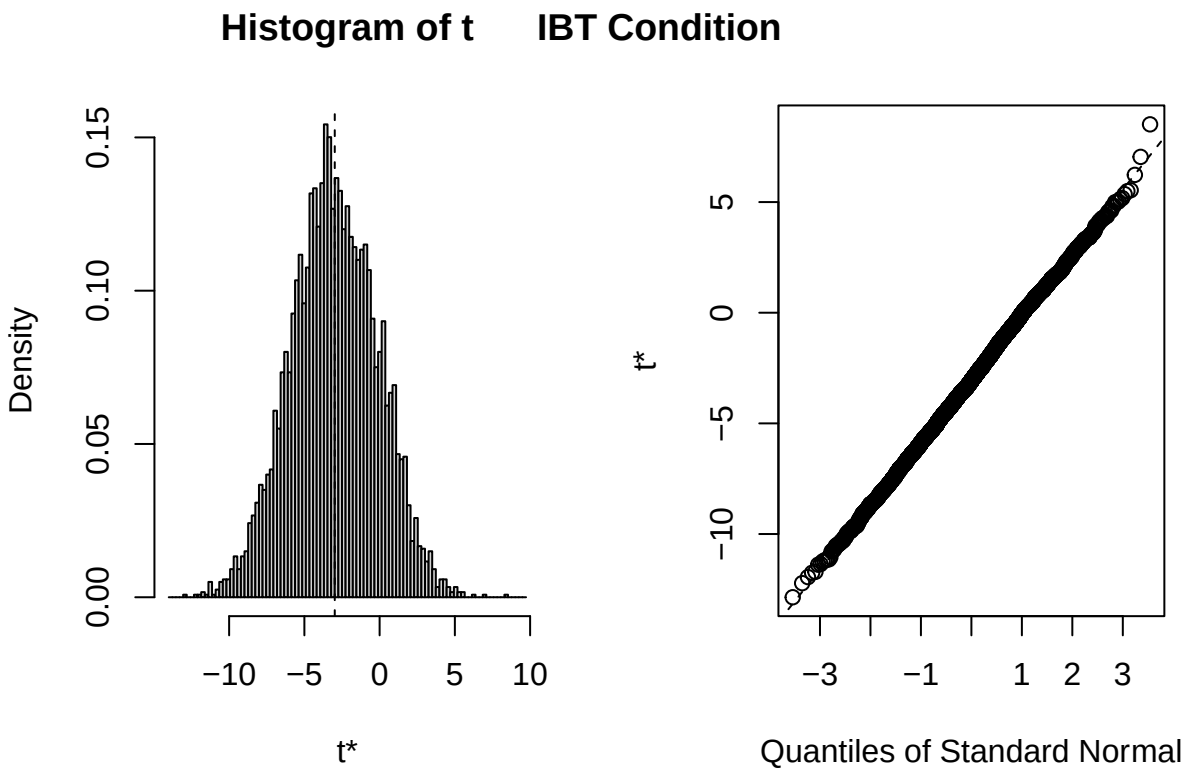
Bootstrap Distribution










Histogram of t IBT Condition
Density
0.00
0.05
0.10
0.15
−10
−5
0
5
10
t*
t*
−10
−5
0
5
−3
−1
1
2
3
Quantiles of Standard Normal